Spokes & Flows: Milestone 2, Data Analysis and Sketches

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GitHub: github.com/HarshithKoriRaj/spokesnflows

1 What changed since Milestone 1

We used the feedback's key question, "where does the system struggle, and when?", to narrow the project. Our main reader is someone planning the Citi Bike network, such as NYC DOT staff deciding where to add docks.

- **Fewer questions.** Old Q2 (where trips happen) and Q3 (imbalance) are merged into one map (new Q2). Old Q7 (new stations as a network) is dropped because it needs a second year of data and doesn't answer the main question. The e-bike question (Q5) is now optional.
- **Q3 refocused.** We planned to focus on East River crossings, but they are only 6.5% of weekday morning trips. Q3 now covers all neighborhood-to-neighborhood flows, with East River crossings as a filter.
- **Imbalance is a trip measure.** Net flow (arrivals − departures per dock) counts riders' trips. It doesn't show whether a station was actually empty or full, because staff rebalancing trips are not in the public data. We will label it that way on the map.
- **Three months.** We use January, July and October 2025 (11,844,428 trips), one month each from winter, summer and fall, instead of the full year.

2 Data handling

Sources (download steps are in the repository README):

- Citi Bike trip histories (Lyft), monthly files from the public S3 bucket. One row is one trip.
- Dock capacity from the Citi Bike GBFS feed.
- 2020 Neighborhood Tabulation Areas (NYC Open Data).
- NOAA daily weather, Central Park.

Processing. `scripts/build.py` loads the trips into DuckDB and:

1. keeps trips that start in the three months and last 1 minute to 3 hours, and drops trips to or from Citi Bike's own depots and test docks;
2. fixes station IDs written two ways (e.g. `7154.1` and `7154.10`), which otherwise split 98 docks in two;
3. adds duration, distance, hour, day of week and weekday/weekend (public holidays count as weekend);
4. joins dock capacity (by station ID, then by location within 75 m), neighborhood and borough, and weather by date;
5. writes one small table per view (Table 2).

Check	Trips	% of raw	What we do
Trips in the downloaded files	11,844,428	100.00	–
Started outside the downloaded months	1,626	0.01	dropped
Duplicate ride IDs	0	0.00	dropped (kept first)
Shorter than 1 min	0	0.00	dropped
Longer than 180 min	16,130	0.14	dropped
Station ID written two ways (7154.1 / 7154.10)	171,362	1.45	ID fixed
To or from a Citi Bike depot or test dock	2,779	0.02	dropped
No start station ID	6,704	0.06	kept; not in station views
No end station ID	37,704	0.32	see below
of which e-bikes	35,368	0.30	
of which classic bikes	2,336	0.02	
with end coordinates	38	0.00	
still in data after cleaning	30,361	0.26	kept; not in station views
Round trips (same start and end)	230,819	1.95	kept
Trips after cleaning	11,824,169	99.83	–
Stations found in trips	2,272		
capacity matched by ID	2,182		
capacity matched by location	30		within 75 m
capacity missing	60		shown as “no data”
outside any NYC neighborhood	66		

Table 1: Data checks. Percentages are of the raw trip count.

Missing data.

- **No end station:** 37,704 trips, mostly e-bikes. Almost none have end coordinates either, so the 2025 files have essentially no trips ending at a known spot away from a dock. The ones that pass the cleaning rules are kept in the time and weather views but left out of the station and neighborhood views.
- **No dock capacity:** 59 NYC stations. They are drawn in grey as “no dock data”.
- **Outside NYC:** 66 Jersey City and Hoboken stations, left off the map.

Processed table	One row is	Rows	Used by
`hourly_profile.csv`	day of week × hour × rider type	336	Q1
`station_hour.csv`	station × weekday/weekend × hour	109,056	Q2
`stations.csv`	station (location, docks, neighborhood)	2,272	Q2, Q3
`od_nta.csv`	neighborhood pair × time period	32,019	Q3
`daily.csv`	day, with weather	93	Q4
`distance_hist.csv`	bike type × 0.25 km distance bin	66	Q5

Table 2: Processed tables and the question each one answers.

3 Domain questions

Q1. When do people ride, and how do members and casual riders differ?

Transformation: count trips by day of week, start hour and rider type, then divide by the number of times that weekday occurs to get trips per day (holidays left out).

day	hour	rider type	trips (study period)	trips per day
Wednesday	8	casual	13,649	975
Wednesday	8	member	145,317	10,380
Wednesday	17	casual	28,211	2,015
Wednesday	17	member	180,777	12,913
Saturday	8	casual	7,544	629
Saturday	8	member	35,775	2,981
Saturday	17	casual	28,019	2,335
Saturday	17	member	80,097	6,675

Table 3: Sample of `hourly_profile.csv` (336 rows in total).

Planned view: two heatmaps side by side, members and casual riders. The x-axis is hour of day and the y-axis is day of week. Shading shows each hour's share of that group's weekly trips, on one shared scale (legend: few to many). Hovering a cell shows the exact count.

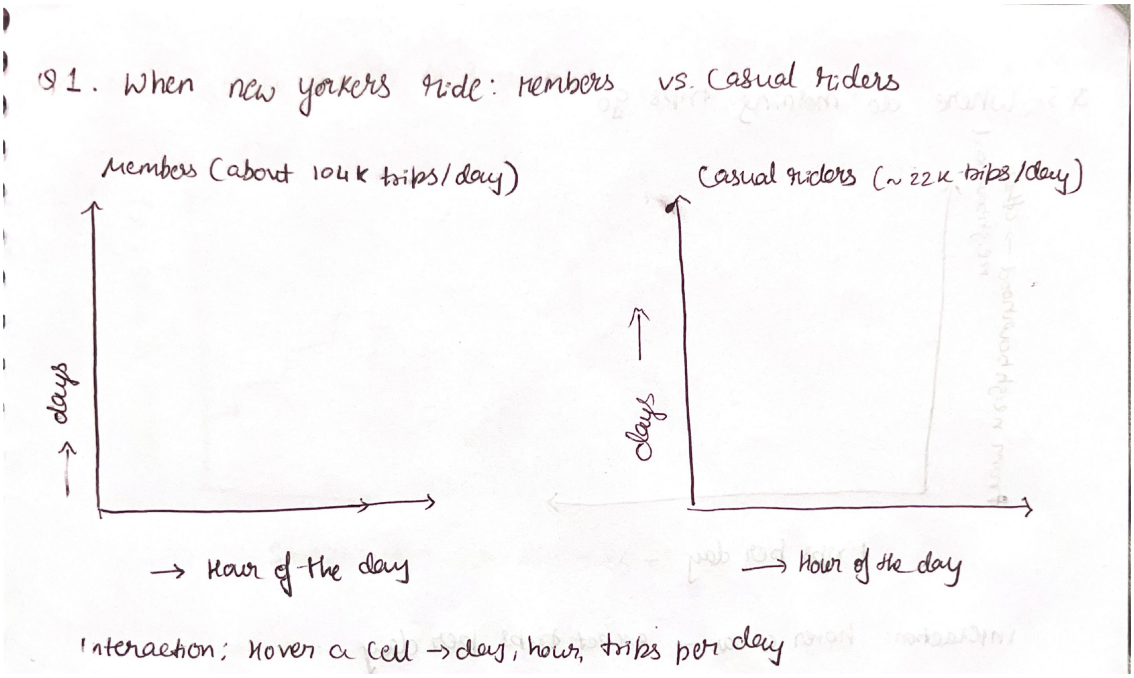


Figure 1: Q1 sketch.

Q2. Where and when do stations gain or lose bikes through trips?

Changed since M1: old Q2 and Q3 merged into one map; the measure is now a trip measure, normalized by docks.

Transformation: for each station, weekday/weekend and hour, count departures and arrivals per day. Net = arrivals − departures, divided by the station's docks, then summed from midnight.

station	hour	dep/day	arr/day	net/day	docks	net/dock	cum. net/dock
W 21 St & 6 Ave	6	10.5	16.3	5.8	74	0.08	0.17
W 21 St & 6 Ave	7	17.5	20.8	3.3	74	0.04	0.21
W 21 St & 6 Ave	8	35.7	49.1	13.4	74	0.18	0.39
W 21 St & 6 Ave	9	32.4	46.9	14.5	74	0.20	0.59
W 21 St & 6 Ave	10	20.3	25.1	4.8	74	0.07	0.65
W 21 St & 6 Ave	11	20.3	22.6	2.3	74	0.03	0.68

Table 4: Sample of `station_hour.csv`: W 21 St & 6 Ave, weekday mornings.

Planned view: a diverging bar chart of stations. The x-axis is net trips per dock since midnight, centered at zero, and the y-axis lists the stations (with borough), ranked. Losing stations are hatched red bars and gaining stations solid blue bars. An hour slider and a weekday/weekend toggle sit above the chart. Clicking a bar shows that station's hourly departures and arrivals.

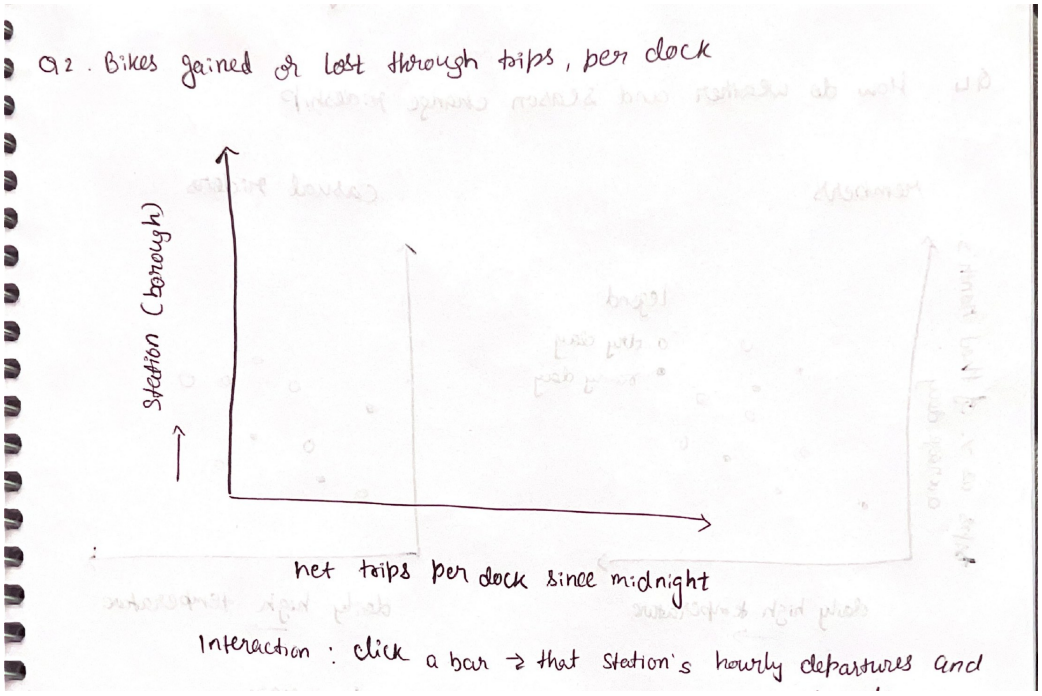


Figure 2: Q2 sketch.

Q3. Where do the trips behind that imbalance come from?

Changed since M1: East River crossings are no longer the focus; they are only 6.5% of weekday morning trips (Table 6).

Transformation: map each trip's start and end station to its neighborhood and borough, then count trips per day for each pair and time period. Pairs that cross a borough line or the East River are flagged.

from neighborhood	to neighborhood	trips/day	East River
Chelsea-Hudson Yards	Midtown South-Flatiron-Union Square	239.6	no
Chelsea-Hudson Yards	Midtown-Times Square	230.9	no
Hell's Kitchen	Midtown-Times Square	227.3	no
Greenpoint	Williamsburg	202.3	no
East Village	Greenwich Village	194.8	no
Financial District-Battery Park City	Tribeca-Civic Center	182.0	no
Murray Hill-Kips Bay	Midtown-Times Square	174.4	no
West Village	Chelsea-Hudson Yards	168.2	no

Table 5: Sample of `od_nta.csv`: busiest pairs of different neighborhoods, weekday 7–10.

period	trips/day	same nbhd %	same borough %	other borough %	East River %
Weekday 7–10	25,129	21.2	69.8	9.0	6.5
Weekday 16–19	35,156	21.1	69.8	9.1	6.6
Weekend (all day)	112,271	21.9	68.4	9.7	6.7

Table 6: Share of trips by destination type and time period.

Planned view: a horizontal bar chart of the busiest neighborhood-to-neighborhood flows. The x-axis is trips per day and the y-axis lists "from → to" pairs. Flows within one borough are solid bars and flows that cross a borough line are dotted bars. A period dropdown switches between weekday morning, evening, other hours and weekend. Hovering a bar shows the exact count.

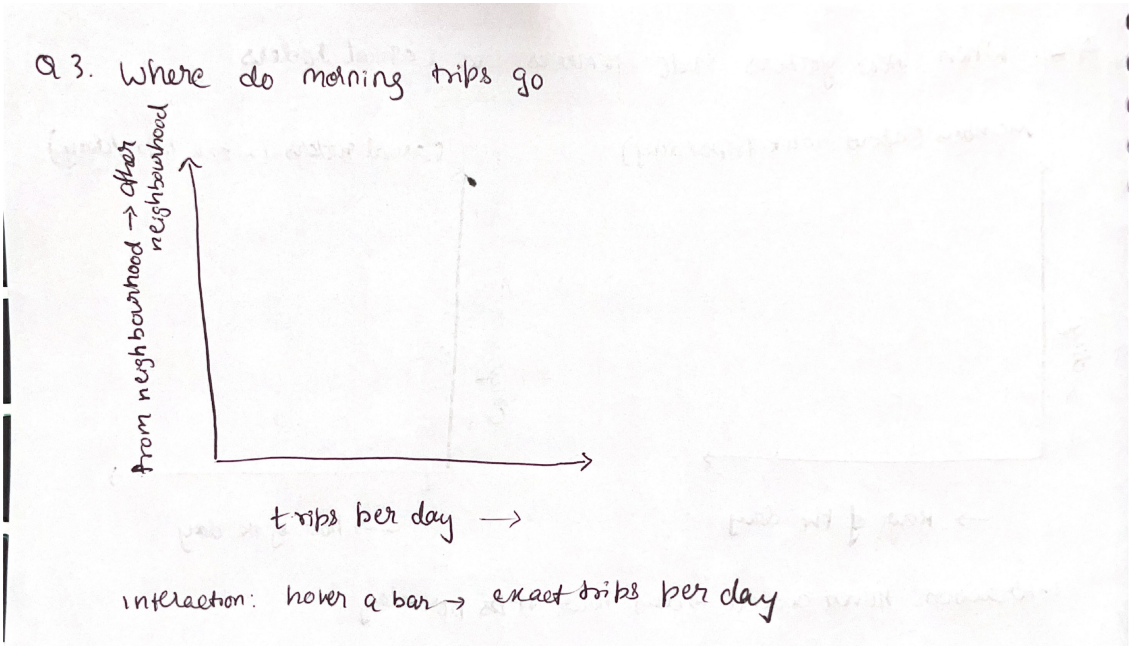


Figure 3: Q3 sketch.

Q4. How much do weather and season change ridership?

Changed since M1: renumbered from old Q6; uses three months instead of the full year.

Transformation: count trips per day by rider type and join that day's temperature, rain and snow by date. Holidays are flagged.

date	day	trips	members	casual	high (F)	rain (in)	snow (in)
2025-01-01	Wednesday	51,363	40,493	10,870	51	0.00	0.00
2025-01-02	Thursday	71,729	63,300	8,429	43	0.00	0.00
2025-01-03	Friday	75,308	66,305	9,003	39	0.00	0.00
2025-01-04	Saturday	55,358	48,148	7,210	33	0.00	0.00
2025-01-05	Sunday	52,147	45,960	6,187	33	0.00	0.00
2025-01-06	Monday	58,483	54,441	4,042	30	0.08	0.90
2025-01-07	Tuesday	65,170	60,754	4,416	33	0.00	0.00
2025-01-08	Wednesday	66,551	62,073	4,478	30	0.00	0.00
2025-01-09	Thursday	67,333	62,557	4,776	33	0.00	0.00
2025-01-10	Friday	81,334	73,493	7,841	39	0.00	0.00

Table 7: Sample of `daily.csv` (93 rows in total).

Planned view: two scatter plots side by side, members and casual riders, with one shared y-axis. The x-axis is the daily high temperature (°F) and the y-axis is trips as a percent of that month's average day. Filled dots are rainy days and hollow dots dry days. Hovering a dot shows its date, and a month filter shows January, July, October or all three.

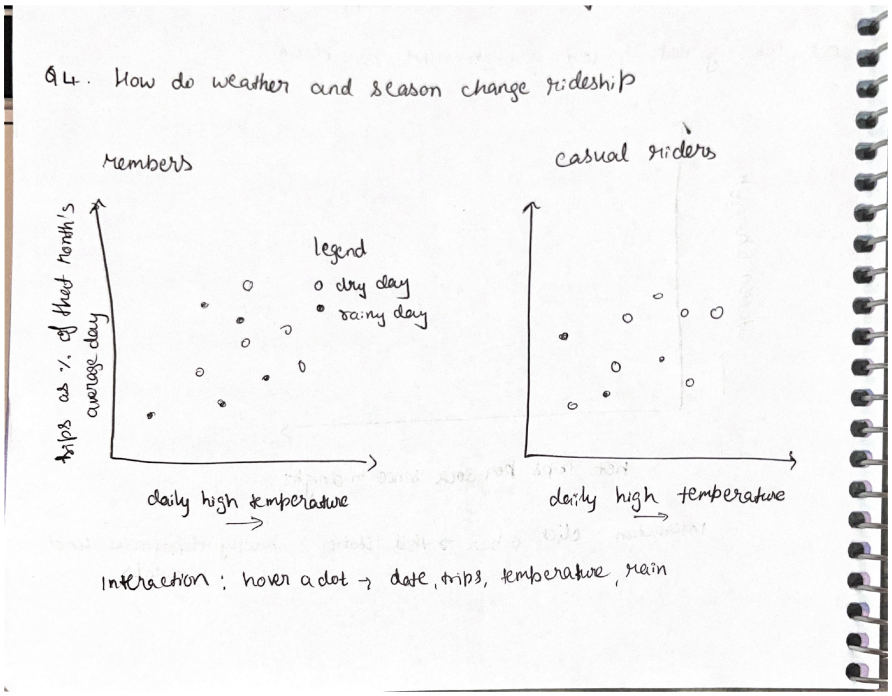


Figure 4: Q4 sketch.

Q5 (optional). Do e-bike trips differ from classic-bike trips?

Changed since M1: now optional.

Transformation: straight-line distance for each trip, grouped into 0.25 km bins by bike type; the last bin holds every trip of 8 km or more.

bike	distance bin	trips	share of that bike's trips %
classic	0.00–0.25 km	207,219	6.0
classic	0.25–0.50 km	406,935	11.7
classic	0.50–0.75 km	488,314	14.1
classic	0.75–1.00 km	449,515	12.9
electric	0.00–0.25 km	197,696	2.4
electric	0.25–0.50 km	367,710	4.4
electric	0.50–0.75 km	713,843	8.6
electric	0.75–1.00 km	872,858	10.5

Table 8: Sample of `distance_hist.csv`.

Planned view: overlaid step histograms. The x-axis is straight-line trip distance (km) and the y-axis is share of trips (%). E-bikes are a solid line and classic bikes a dashed line, with each median marked. Hovering a bin shows its share.

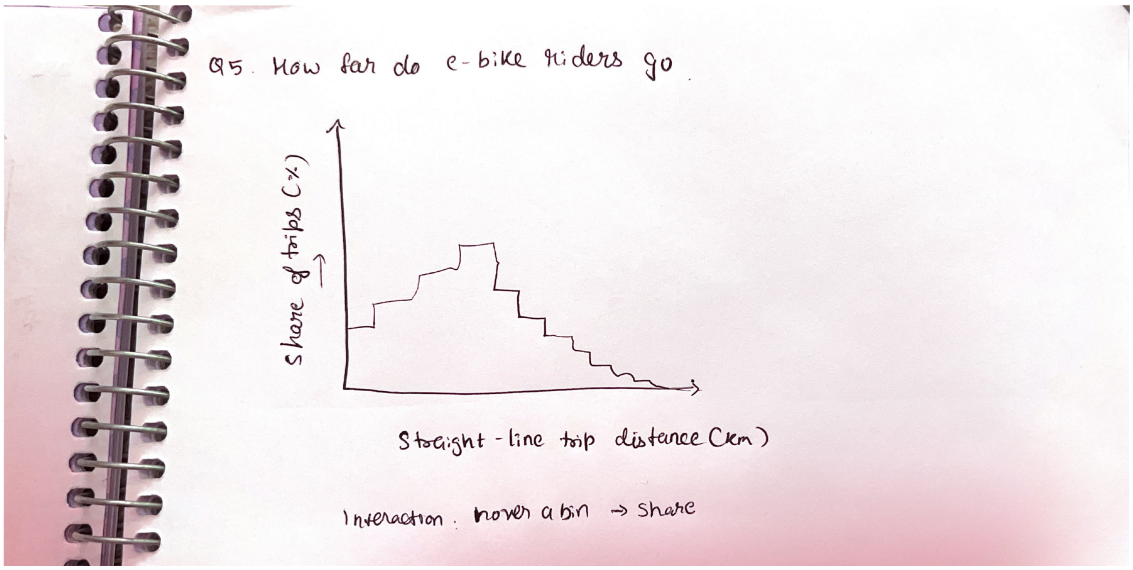


Figure 5: Q5 sketch.

AI use. We used Claude (Anthropic) to help write the data scripts, check the data and draft parts of this report, and to make reference mockups from our processed data. We drew the sketches by hand.